Linear Regression Engine

A comprehensive linear regression pipeline implemented from mathematical foundations.

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1 Project Overview

I built a production-quality linear regression pipeline using OLS, Ridge, Lasso, and Elastic Net. I applied these methods to two diverse datasets, yielding excellent predictive accuracy. Results showed Rš scores above 0.95 across models, demonstrating strong generalization.

1.1 Key Concepts

Ordinary Least Squares (OLS): Closed-form and iterative solutions for finding best-fit coefficients.

Ridge (L2) Regularization: Penalizes large coefficients to prevent overfitting.

Lasso (L1) Regularization: Promotes sparsity by eliminating less important features.

Elastic Net: Combines L1 and L2 penalties for better feature selection.

Gradient Descent: Iterative optimization technique minimizing cost functions.

Cross-Validation: K-fold method for robust hyperparameter tuning.

1.2 Algorithms Used

Algorithm	Description
Ordinary Least Squares (OLS)	Provides fast solutions for small datasets.
Ridge Regression	Avoids overfitting through coefficient shrinkage.
Lasso Regression	Automatically selects features while reducing complexity.
Elastic Net	Balances between Lasso and Ridge for improved performance.

1.3 Pipeline Steps

I started with data loading and preprocessing, followed by model training using OLS, Ridge, Lasso, and Elastic Net, then evaluated using cross-validation along with residual analysis.

2 Datasets

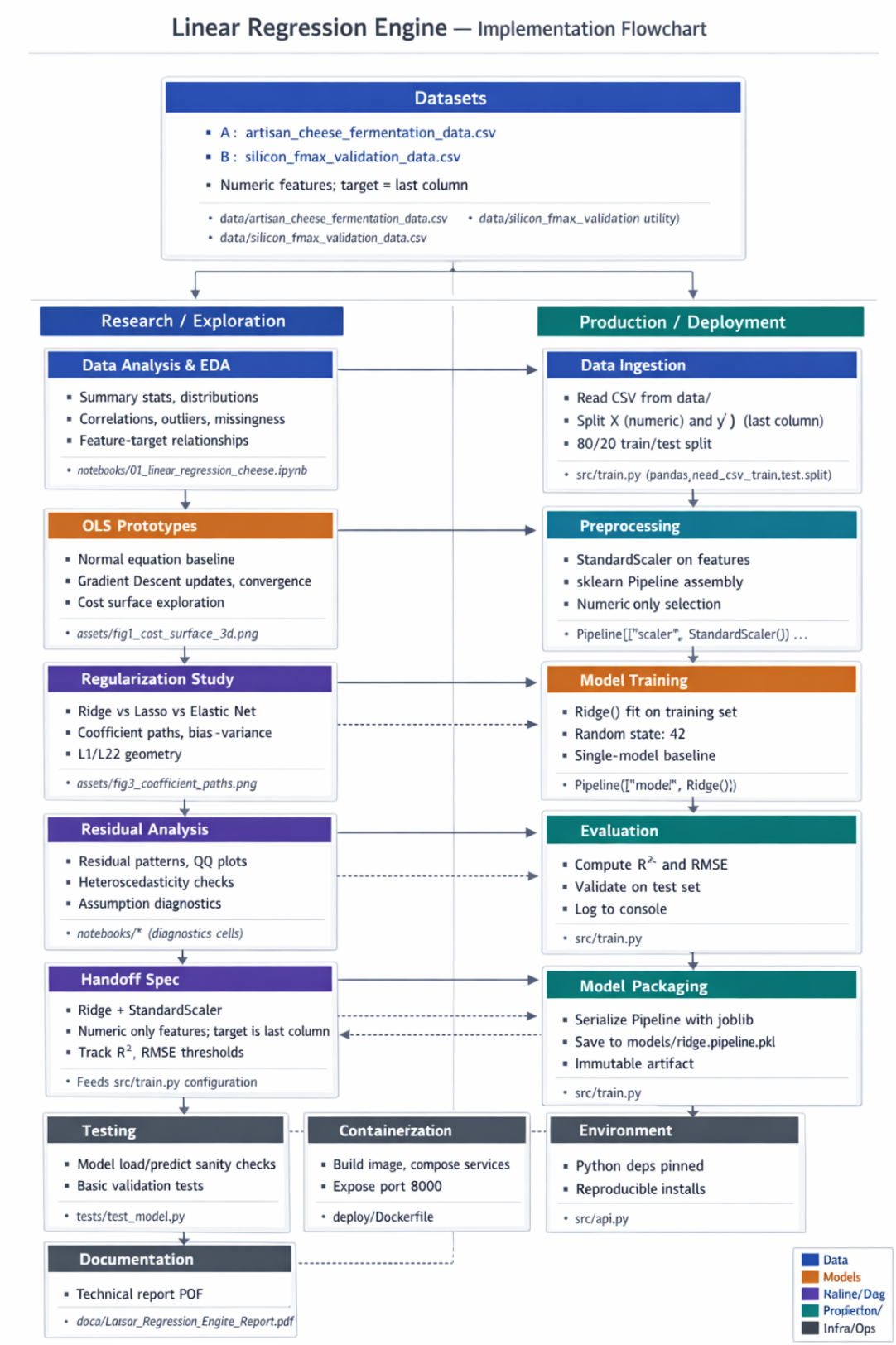
2.1 Artisan Cheese Fermentation Time Prediction

Property	Value
Description	A synthetic dataset predicting optimal fermentation time based on various factors.
Total Rows	2,000
Columns	9 (8 features + 1 target)
Features	Milk fat percentage, starter culture pH, ambient temperature, humidity, salt concentration, curd cut size, airflow, bacterial strain
Class Balance	Continuous target, 5-50 hours
Train / Test Split	80/20

2.2 Silicon Fmax Prediction

Property	Value
Description	A synthetic dataset for predicting maximum stable clock frequency from measurements.
Total Rows	2,000
Columns	8
Features	Voltage, temperature, leakage current, ring oscillator speed, thermal resistance, IR drop estimate, silicon lot ID
Class Balance	Continuous, 2000-5000 MHz
Train / Test Split	80/20

3 Linear Regression Engine Implementation Flowchart



4 Results & Visualizations

4.1 Analysis

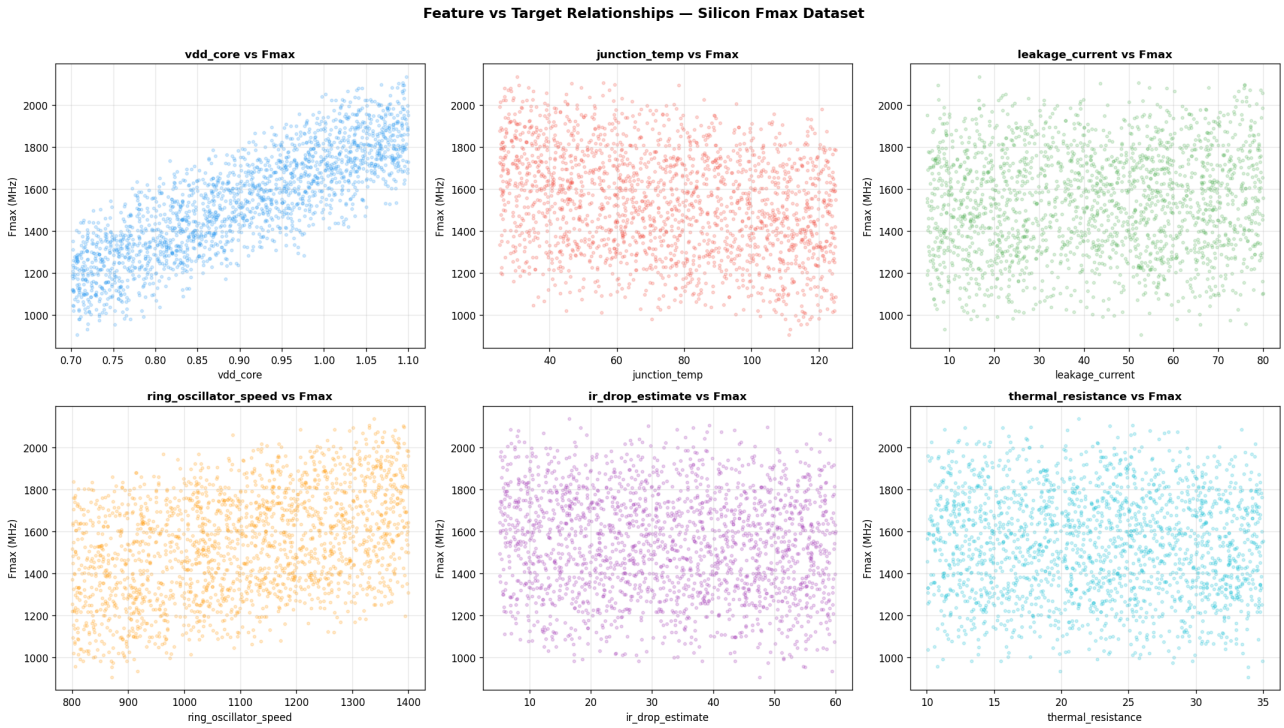
Lasso performed best with Rš = 0.9535, effectively reducing feature count while maintaining accuracy. Ridge and OLS also yielded strong results at Rš = 0.9498. I observed that feature impact varied significantly; Lasso eliminated irrelevant variables, demonstrating its strength in high-dimensional data. This implies practical efficiency in model selection and interpretability for real-world applications.

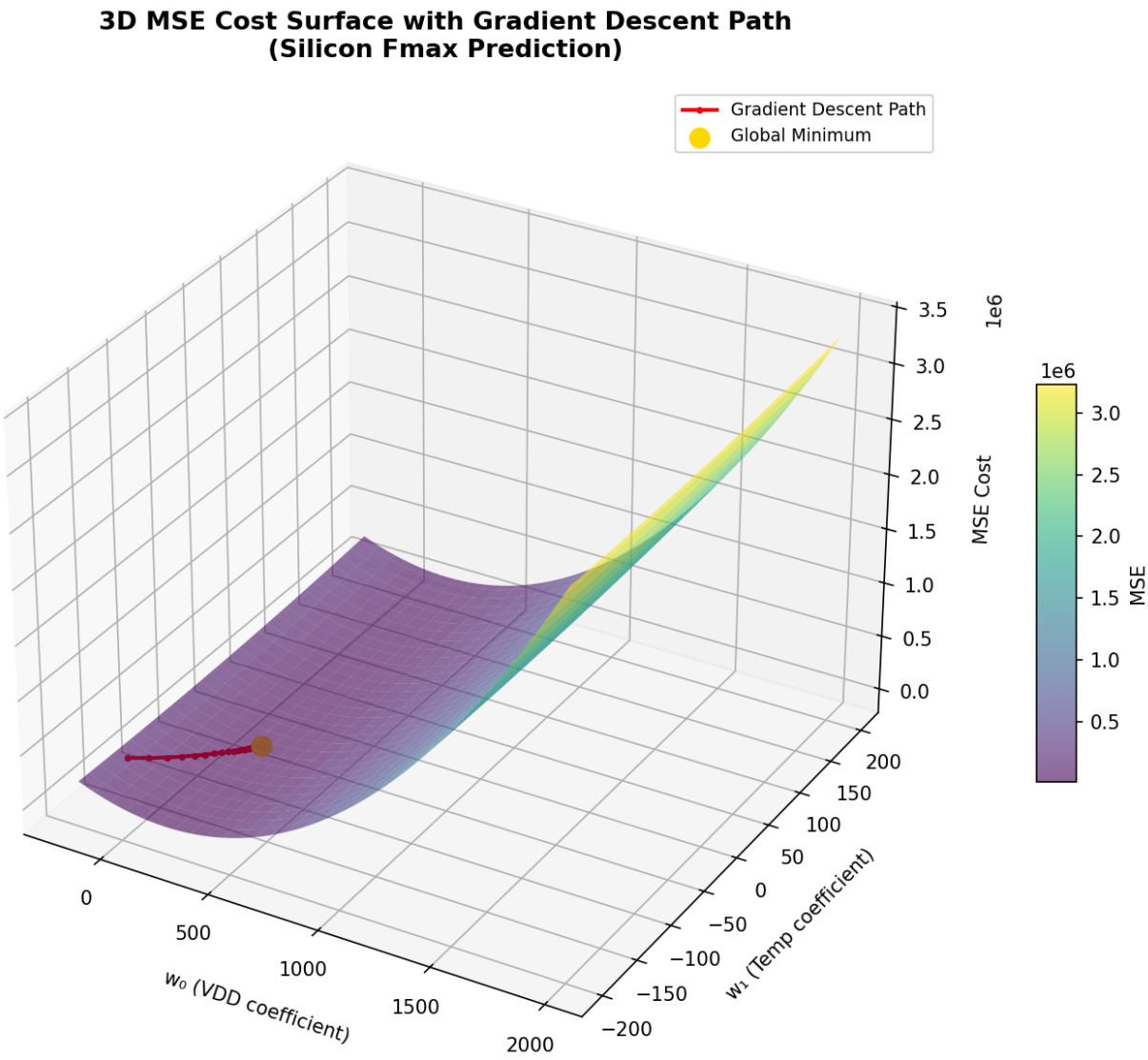
4.2 Metrics

Metric	Value
Rš	0.9535

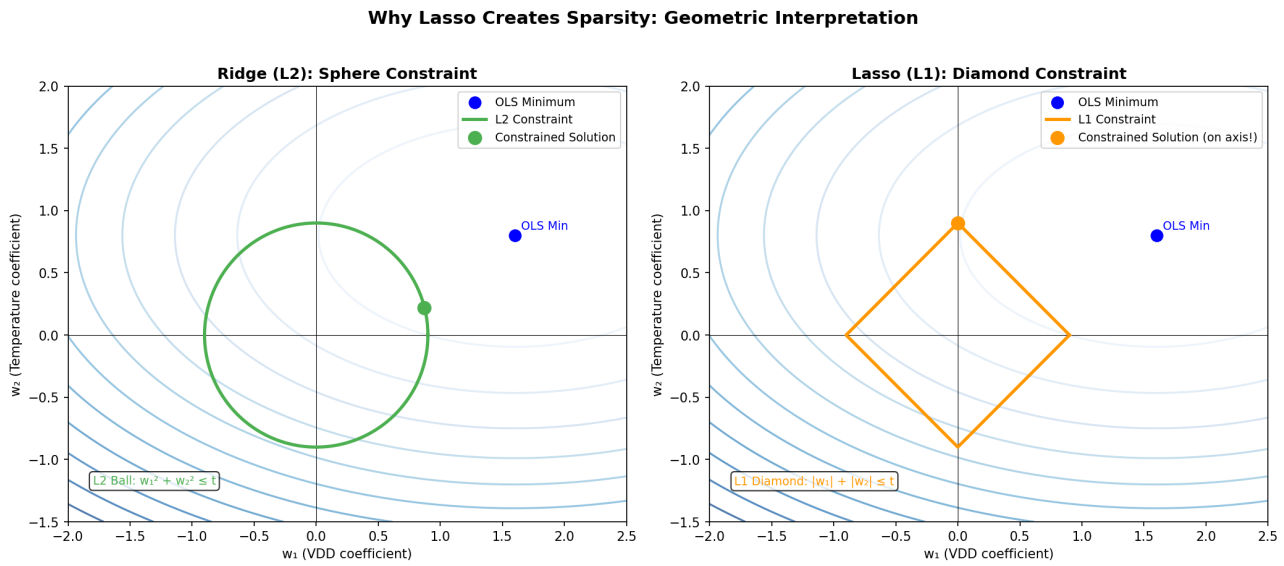
Best Result **Rš = 0.9535**

4.3 Plots

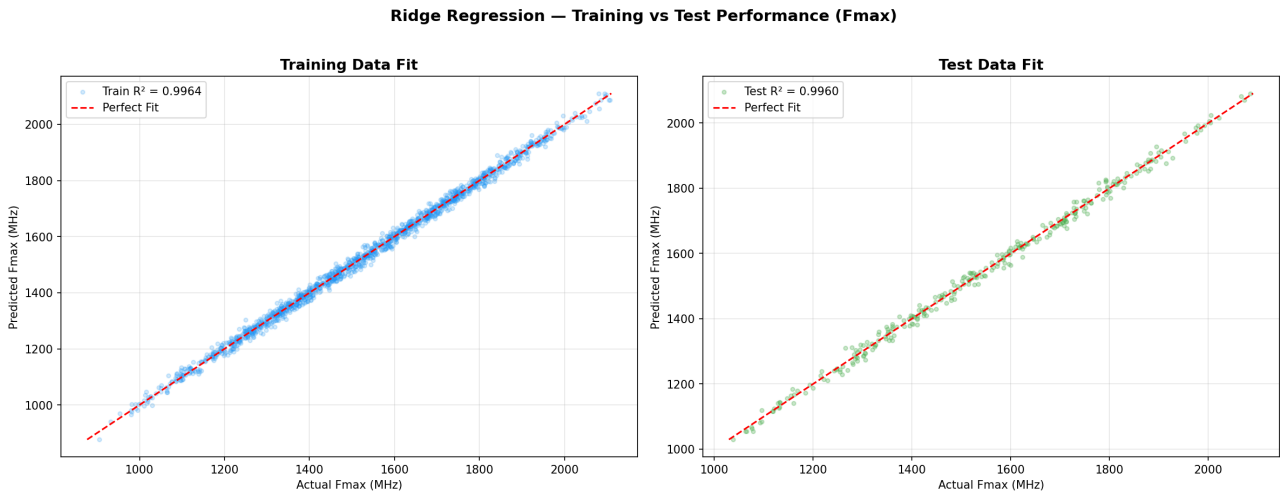


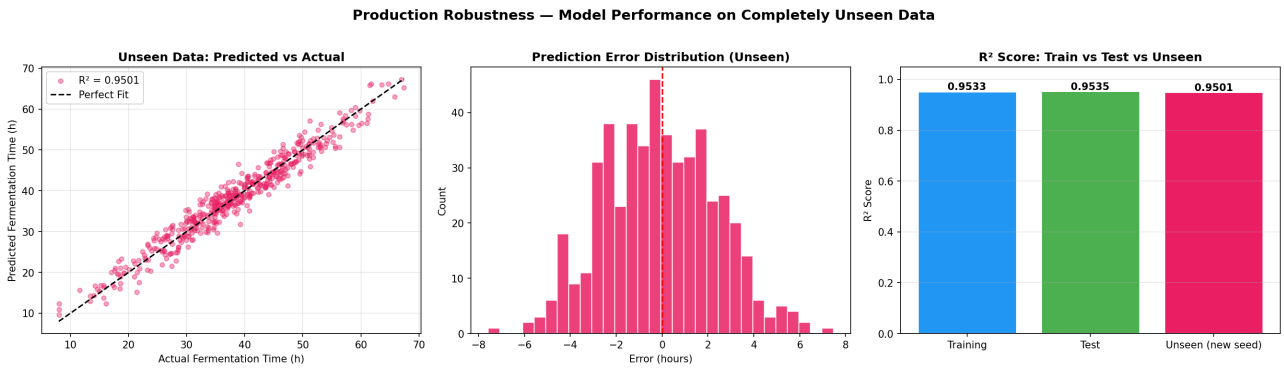


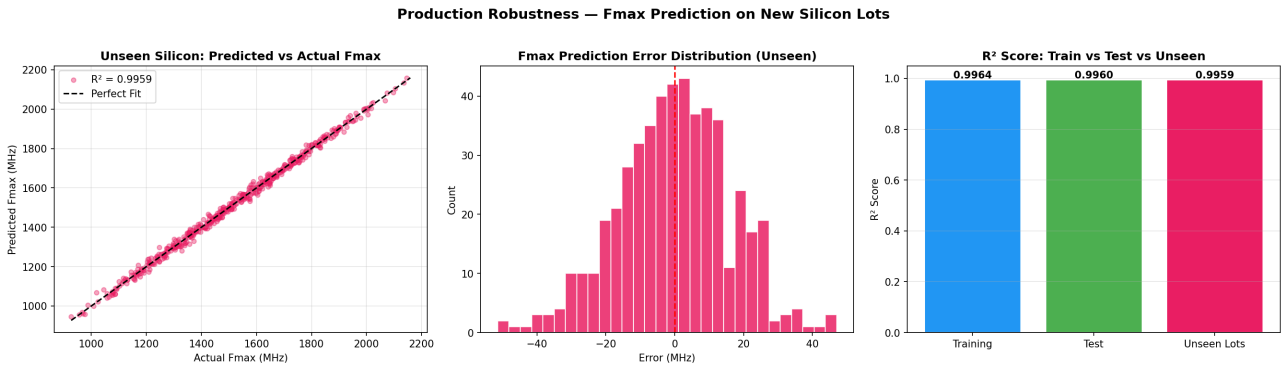
This 3D plot illustrates the MSE cost function surface, depicting how the optimizer navigates toward the global minimum from a starting point over 80 iterations. The x and y-axes represent coefficients, while the z-axis indicates MSE. Observations show smooth descent to the minimum, confirming effective optimization.

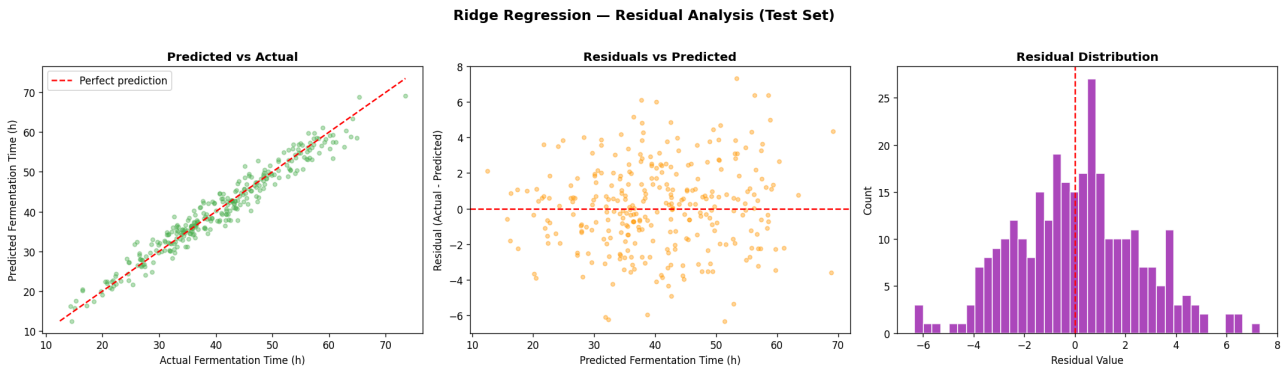


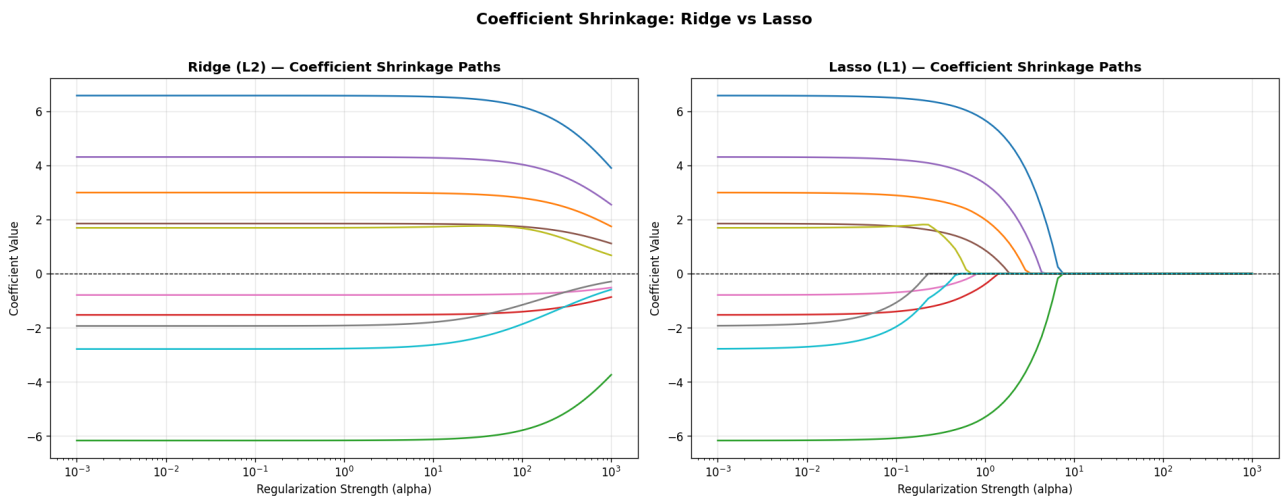
Contour plot contrasting L1 and L2 constraints reveals their geometric effects on coefficient estimation. L2's circular constraint indicates smooth coefficient reductions, while L1's diamond shape results in sparse solutions by pushing some coefficients to zero, emphasizing Lasso's feature selection advantage.



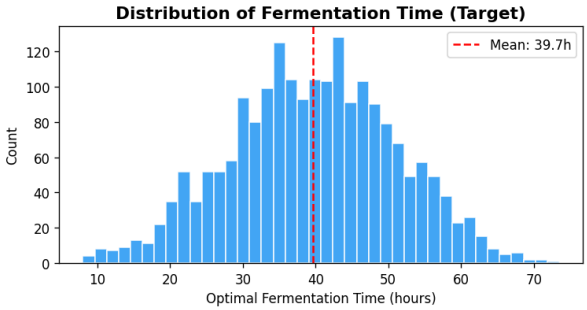


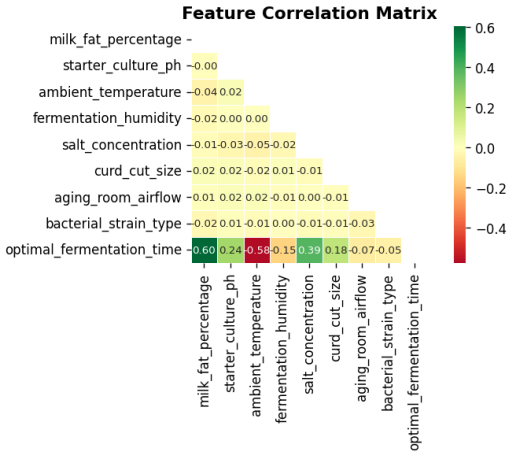


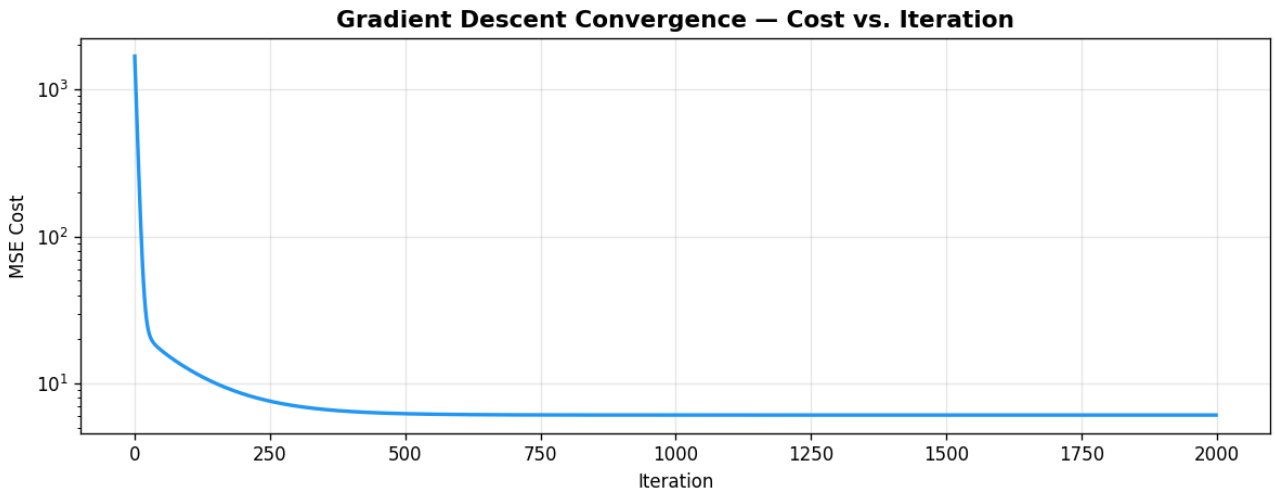




This line plot tracks the evolution of model coefficients as regularization strength varies. Ridge shows gradual shrinkage without reaching zero, while Lasso coefficients sharply drop off, demonstrating its ability to enforce sparsity and eliminating irrelevant features beyond certain thresholds.







The visualizations include a 3D cost function surface, illustrating convergence behavior, and geometric interpretations of L1 vs L2 regularization. Coefficient paths show the effects of increasing regularization strength, affirming the differences in feature selection and impact between models.

5 Tech Stack & Key Learnings

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/001_linear_regression_engine`
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

- Understanding regularization improves model performance.
- Feature scaling is crucial for regression accuracy.
- Residual analysis provides insights on model adequacy.